Macroeconomics PhD Intro
Lecture 1: Measurement, History, and Stylized Facts

Based on Part 1 of course book material

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Lecture Roadmap

- Why macroeconomics is empirical and policy-relevant.
- How national accounts and price indices create the object we model.
- Historical episodes that changed theory and method.
- Stylized facts from Chapter 2 that discipline all macro models.
- Bridge to Lecture 2: the Solow model.

What Is Macroeconomics?

- Study of aggregates: output, consumption, investment, employment, inflation.
- Study of distributions: income, wealth, and heterogeneity across households and firms.
- Study of dynamic general equilibrium with quantitative discipline.
- Study of current events through theory tied to measurement.

Organizing Question

Core Question

What is going on in the macroeconomy, and what can policy do about it?

- Long run: growth, productivity, structural change.
- Medium run: transitions, convergence, reforms.
- Short run: recessions, financial stress, inflation episodes.

What Do We Measure?

- National accounts begin in nominal values (current prices).
- Real quantities require deflation by a general price index.
- Measurement choices are not neutral: they encode theory.

Takeaway

Before explaining output fluctuations, we must define output correctly.

Price Index Construction

- A price index is a weighted average of prices.
- Weights can be fixed (base-year) or time-varying (chain-weighted).
- Functional form matters: arithmetic, geometric, or superlative indices.

Practical Point

Different index formulas can change measured real growth.

Appendix

National Income and Product Accounts

Three Equivalent GDP Views

$$Y = C + I + G + NX = \text{Labor Income} + \text{Capital Income} = \sum \text{Value Added}_i$$

- Expenditure side, income side, and production side must match.
- Value-added accounting avoids double counting of intermediates.

Appendix

GDP Is Not Welfare

- GDP measures market production, not total well-being.
- It omits leisure, many environmental amenities, and household production.
- Welfare analysis requires additional objects beyond GDP.

Measurement Under Stress: COVID Episode

- Pandemic shock forced near real-time macro measurement.
- High-frequency transactions (cards, mobility, payroll) became key inputs.
- Empirical macro expanded toward administrative and private data.

Great Depression and the Birth of Modern Macro

- Depression raised the diagnostic question: what is going on?
- Systematic aggregate data made theory testable.
- Keynesian analysis emphasized demand shortfalls with nominal rigidities.

Long-Run Growth Revolution

- Solow (1956, 1957): growth theory + growth accounting.
- Output growth decomposed into capital, labor, and residual productivity.
- The residual (TFP) became central for interpreting long-run growth.

Appendix

1970s Stagflation and Policy Credibility

- High inflation and unemployment challenged static Phillips-curve policy.
- Expectations and policy credibility became first-order objects.
- Shift from reduced-form stabilization to explicit dynamic policy design.

Kydland-Prescott: A Way Forward

- Rules vs discretion: commitment is a central policy issue.
- Quantitative macro: disciplined models calibrated to data.
- Dynamic stochastic general equilibrium became a workhorse platform.

Waves of Macroeconometrics

- Structural estimation and calibration of dynamic models.
- VAR-based identification of shocks and propagation.
- Integration of micro and administrative data in macro analysis.

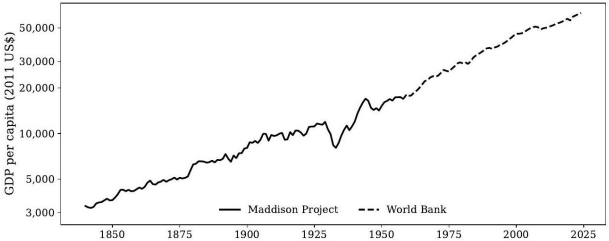
Inequality and Heterogeneity

- Macro now models full distributions, not only representative means.
- Wealth and income heterogeneity change aggregate responses.
- Policy multipliers depend on distributional structure (e.g., MPC dispersion).

Macro Frontier Today

- Global linkages: trade, exchange rates, external balance sheets.
- Climate and energy: transition dynamics and policy tradeoffs.
- Public finance and monetary-fiscal interaction under uncertainty.

Fact 1: Trend Growth in GDP per Capita

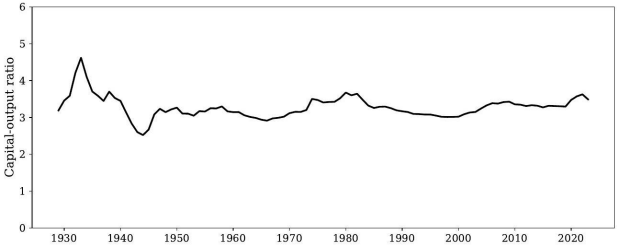


Fact: log GDP per capita is close to linear over long horizons.

Interpretation: sustained trend growth with episodic breaks.

Model implication: growth models must deliver persistent productivity growth.

Fact 2: Capital-Output Stability

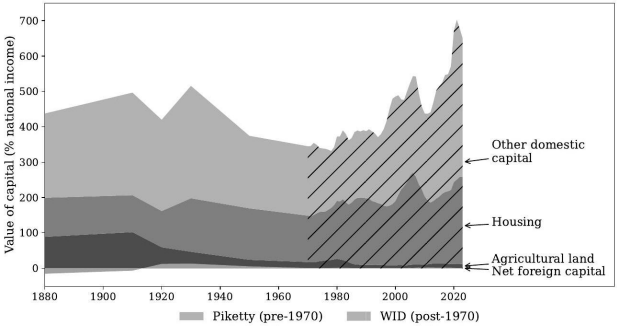


Fact: the capital-output ratio is fairly stable over long spans.

Interpretation: accumulation and depreciation balance in the long run.

Model implication: balanced-growth discipline for production and savings behavior.

Fact 2b: Wealth-Output Dynamics

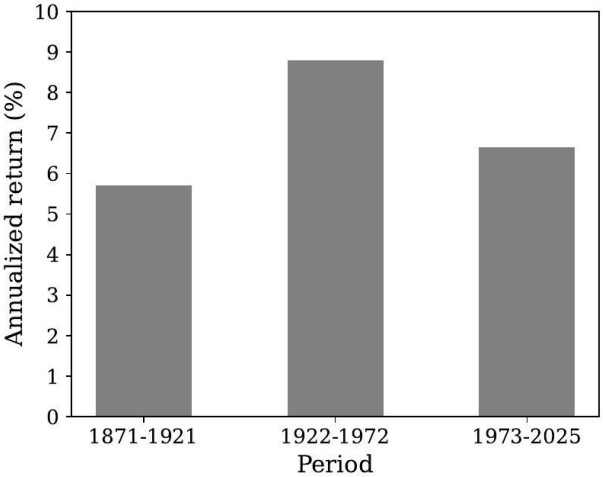


Fact: wealth-output can deviate from capital-output.

Interpretation: valuation, land, and housing matter beyond productive capital.

Model implication: asset-market structure matters for distribution and policy.

Fact 3: Return on Capital

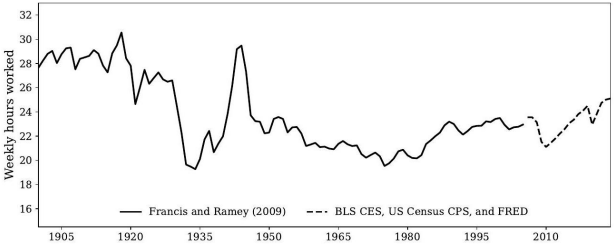


Fact: no dramatic long-run trend in average returns.

Interpretation: user cost is not collapsing over the sample.

Model implication: steady-state calibration with stable capital shares is plausible.

Fact 4: Labor Input Trends

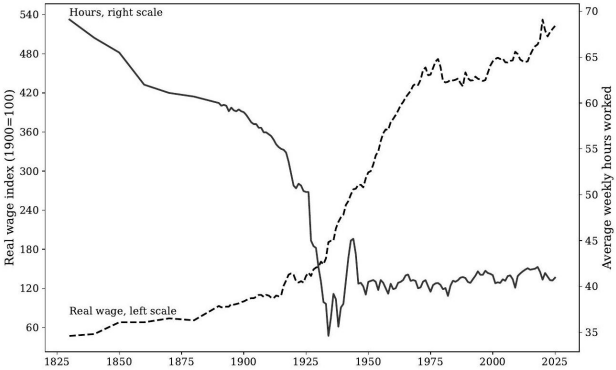


Fact: hours per person show long-run decline and cyclical movement.

Interpretation: participation, employment, and intensive margin all matter.

Model implication: labor supply and labor market frictions are essential.

Fact 5: Hours and Wage Growth

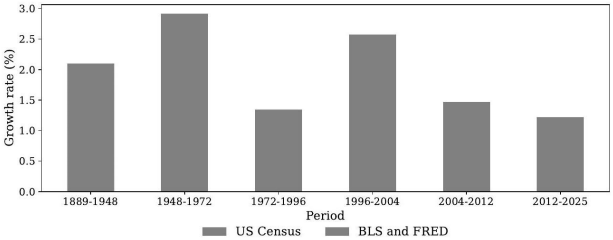


Fact: hours trend down while real wages trend up over long horizons.

Interpretation: income and substitution effects jointly shape labor supply.

Model implication: preference specification matters for balanced-growth hours.

Fact 6: Labor Productivity Growth

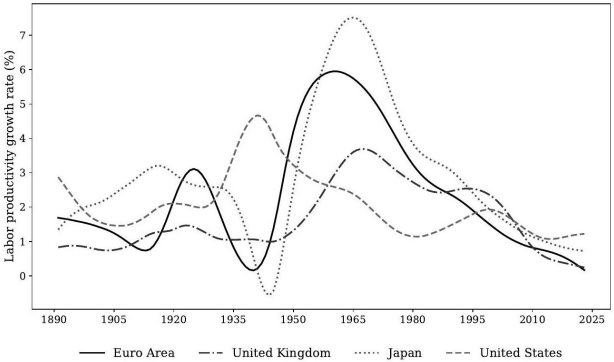


Fact: productivity grows with persistent medium-run variation.

Interpretation: trend and cyclical components both matter.

Model implication: shocks + propagation mechanisms are required.

Fact 6b: Cross-Country Productivity

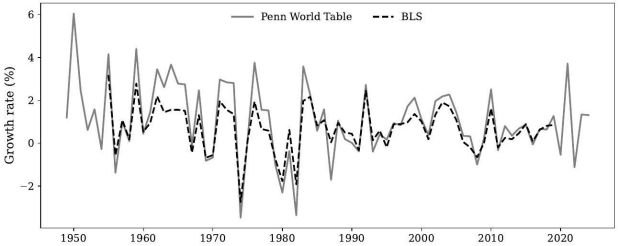


Fact: countries share trend growth but differ in levels and timing.

Interpretation: institutions, technology adoption, and policy differ.

Model implication: convergence is conditional, not automatic.

Fact 6c: TFP Measures

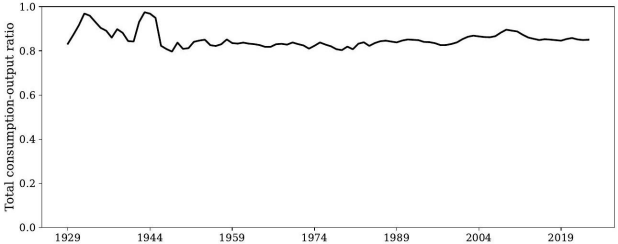


Fact: TFP growth is positive but volatile.

Interpretation: productivity is a key source of macro fluctuations.

Model implication: technology process specification is central in DSGE work.

Fact 7: Expenditure Shares

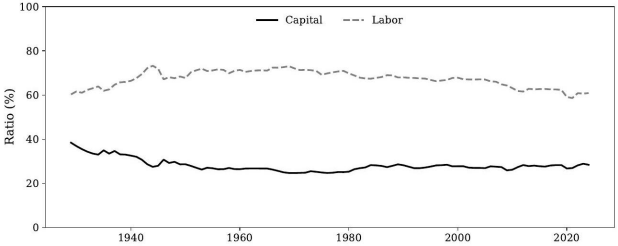


Fact: consumption-output ratio moves over time.

Interpretation: saving and investment shares are not constants year by year.

Model implication: transition dynamics are quantitatively important.

Fact 8: Factor Shares in the U.S.

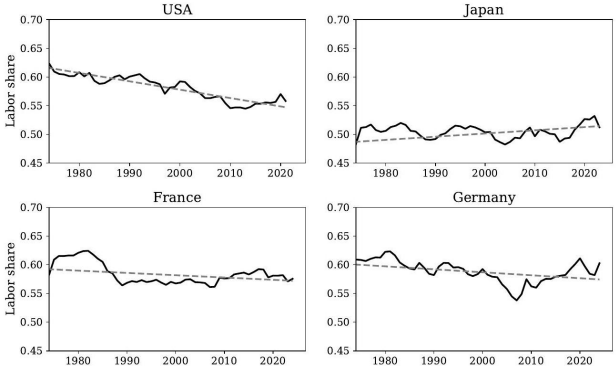


Fact: capital and labor shares are broadly stable over long spans.

Interpretation: Cobb-Douglas benchmark remains informative.

Model implication: use share-based calibration as a first pass.

Fact 8b: Labor Share Drift



Fact: labor share declines in recent decades in many economies.

Interpretation: markups, technology, and composition effects are candidates.

Model implication: constant-share assumptions require robustness checks.

Global Labor Share

Fact: labor share decline appears in global aggregates too.

Interpretation: structural forces are broad, not purely country-specific.

Model implication: open-economy and firm-heterogeneity channels matter.

Synthesis

Workflow for the Course

Measurement $\rightarrow$ Stylized Facts $\rightarrow$ Theory $\rightarrow$ Quantitative Evaluation $\rightarrow$ Policy

- Macro is a cumulative conversation between data and theory.
- Models are judged by disciplined fit to moments and mechanisms.
- Policy advice inherits the strengths and limits of measurement.

Bridge to Lecture 2: Solow Model

- Why can K/Y remain stable while output grows over time?
- What drives long-run growth in income per capita?
- Do poorer economies converge to richer ones, and under which conditions?

Next Lecture

We formalize these questions in Chapter 3 using the Solow model and its dynamic system.

A1. CES and Tornqvist Price Index Intuition

- A superlative index approximates true cost-of-living changes better than fixed-base Laspeyres.
- Tornqvist uses average expenditure shares and log price changes:

$$\Delta \ln P_t \approx \sum_i \bar{s}_{i,t} \Delta \ln p_{i,t}.$$

- In CES settings, exact price indices have explicit aggregator forms.

Back

A2. Value-Added Accounting Example

- Firm A sells unfinished car body for 20; Firm B paints and sells final car for 30.
- Gross sales sum to 50, but GDP contribution is only value added: $20 + (30 - 20) = 30$.
- Value-added accounting prevents counting intermediate transactions twice.

Back

A3. Growth Accounting in Continuous Time

- Start from $Y = F(K, L, t)$ and totally differentiate.
- Under competitive pricing and CRS:

$$\frac{\dot{Y}}{Y} = \alpha \frac{\dot{K}}{K} + (1 - \alpha) \frac{\dot{L}}{L} + \text{TFP growth.}$$

- TFP growth is the residual after accounting for measured input growth.

Back

A4. Why Calibration?

- Choose functional forms and parameters disciplined by micro evidence.
- Match long-run macro moments (shares, growth rates, volatility profiles).
- Then evaluate model mechanisms and policy counterfactuals quantitatively.

A5. Data Caveats for Macro Work

- PPP versus market exchange-rate comparisons.
- Chain-weight revisions can alter historical growth decompositions.
- Labor-share and capital-income definitions vary across datasets.